

# Question Paper

Exam Date & Time: 03-Jan-2026 (09:30 AM - 12:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026

SUBJECT: MAT 2122/MAT\_2122 ENGINEERING MATHEMATICS - III

(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

ENGINEERING MATHEMATICS - III [MAT 2122]

Marks: 50

Duration: 180 mins.

Answer all the questions.

1A) Find the Fourier series expansion of the function (3)

$$f(x) = x - x^2, -1 \leq x \leq 1, f(x+2) = f(x).$$

Course Outcome: "CO1" / Cognitive Level: "Application"

1B) Obtain the half-range Fourier cosine series (3)

expansion of  $f(x) = x(\pi - x), 0 \leq x \leq \pi$ .

Course Outcome: "CO1" / Cognitive Level: "Application"

1C) Find the constant term and first sine and cosine terms of (4)  
the Fourier series expansion of  $y$  from the following data.

|     |   |    |    |    |    |    |
|-----|---|----|----|----|----|----|
| $x$ | 0 | 1  | 2  | 3  | 4  | 5  |
| $y$ | 9 | 18 | 24 | 28 | 26 | 20 |

Course Outcome: "CO1" / Cognitive Level: "Application"

2A) Find the Fourier sine transform of  $f(x) = xe^{-3x}$ . (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

2B) Find the homogeneous system whose solution space (3)  
 $W$  is spanned by the set of vectors  
 $\{(2,1,0,1), (1,3,2,1), (0,1,1,1)\}$ .

Course Outcome: "CO3" / Cognitive Level: "Application"

2C) (4)

Find an orthogonal matrix  $P$  that diagonalizes the matrix  $A = \begin{bmatrix} 2 & -2 \\ -2 & 5 \end{bmatrix}$ .

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Solve the system of equations  $x + y - z = 4$ ,  
 $x - 2y + 3z = -6$ ,  $2x + 3y + z = 7$  by the LU  
 decomposition method. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3B) Find the Cholesky decomposition of the matrix  
 $A = \begin{bmatrix} 4 & 2 & 6 \\ 2 & 10 & 9 \\ 6 & 9 & 14 \end{bmatrix}$ . (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3C) Find a singular value decomposition of the matrix  
 $A = \begin{bmatrix} -4 & -7 \\ 1 & 4 \end{bmatrix}$ . (4)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 4A) Let  $W$  be the subspace of  $\mathbb{R}^4$  spanned by the set of vectors  
 $\{(1,1,2,4), (2,-1,-5,2), (1,-1,-4,0), (2,1,1,6)\}$ . Find a  
 basis and the dimension of  $W$ . (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4B) Consider the following subspaces of  $\mathbb{R}^3$ .  
 $U = \text{Span}\{(1,0,1), (0,1,1)\}$  and  
 $W = \text{Span}\{(1,1,0), (1,2,1)\}$ . Find the dimension of  
 $U + W$  and  $U \cap W$ . (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4C) Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$  be a linear transformation defined by  
 $T(x,y,z) = (x - y - 2z, -x + 2y + 3z)$ . Find a basis and  
 the dimension of  
 (i) Kernel of  $T$   
 (ii) Image of  $T$ . (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) (3)

Find the change of basis matrix  $P$  from  $B = \{(1,2,1), (2,3,3), (3,7,1)\}$  to the basis  $B' = \{(3,1,4), (5,2,1), (1,1,-6)\}$  in  $\mathbb{R}^3$ . Hence write down the change of basis matrix from  $B'$  to  $B$ .

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) Let  $V$  be the vector space of all the polynomials over  $\mathbb{R}$  with the inner product  $\langle p(x), q(x) \rangle = \int_a^b p(x)q(x)dx$ . Determine the cosine of the angle between the polynomials  $p(x) = x^3 - x$  and  $q(x) = x^2 + x$  in  $[-1,1]$ . (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) Find an orthonormal basis for the orthogonal complement of the vector  $u = (1,1,0,1)$ . (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

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| Event Name                        | Subject                              |
|-----------------------------------|--------------------------------------|
| 058-MIT BLR-MAKEUP EXAM-2025-2026 | MAT_2122:ENGINEERING MATHEMATICS-III |

| #     | Course Outcome | Description | Marks |
|-------|----------------|-------------|-------|
| 1     | 1              | CO 1        | 13    |
| 2     | 2              | CO 2        | 11    |
| 3     | 3              | CO 3        | 12    |
| 4     | 4              | CO 4        | 7     |
| 5     | 5              | CO 5        | 7     |
| Total |                |             | 50    |

\* **NA** - CO Not Selected

| Event Name                        | Subject                              |
|-----------------------------------|--------------------------------------|
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| #     | Level       | No of Questions | Marks (%) |
|-------|-------------|-----------------|-----------|
| 1     | Remember    | 0               | 0(0%)     |
| 2     | Understand  | 0               | 0(0%)     |
| 3     | Application | 13              | 43(86%)   |
| 4     | Analysis    | 2               | 7(14%)    |
| 5     | Create      | 0               | 0(0%)     |
| 6     | Evaluate    | 0               | 0(0%)     |
| Total |             | 15              | 50 (100%) |